

# David Pérez Arbués

Engineering Lead · adidas Digital Experiences (DXP)

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## ■ SUMMARY

Software engineering leader with 15+ years of experience, currently working within adidas DXP — the engineering organisation that builds and operates adidas.com, the Flagship app, CONFIRMED, and Running. Leading the .com Membership chapter (9 engineers, 5 direct reports), with scope recently expanded to include .com Acquisition (3 engineers focused on Glass decommission) and cross-platform coordination across Upper Funnel surfaces — PDP, PLP, and checkout — through the Latte rendering initiative.

In 2025, stepped up to cover the Director of Software Engineering role during a period of leave — gaining direct exposure to Director-level decision-making and contributing to the 2026 value-led team structure.

Strong hands-on engineering instincts alongside the leadership breadth: in 2026 led AI-enabled engineering transformation — pioneering GenAI-assisted delivery that halved Glass Decommission timelines and earning the adidas Bolder Award (April 2026). A consistent track record of hands-on delivery: critical microfrontend migrations under hard deadlines, internal tooling built before the ecosystem caught up. Grounded in AI/ML research since 2009 (Deep Belief Networks applied to brain-computer interfaces, University of Zaragoza) — a foundation that informs a practical, first-principles approach to AI adoption.

## ■ EXPERIENCE

### Senior Software Engineer → Engineering Lead, .com Membership · adidas DXP

Aug 2022 – present

adidas · Zaragoza, Spain

- 2025 — Covered the Director of Software Engineering role during a period of leave: participated in Director-level meetings with the Senior Director of SE, reported across all of Membership (.com + apps), and collaborated on the 2026 value-led team structure (cross-functional squads: backend, .com, iOS, Android)
- Led .com Membership chapter (9 engineers, 5 direct reports) with expanded scope covering .com Acquisition (3 engineers, Glass decommission); responsible for KLO/RUN/Support — including security issue resolution — coaching, capacity planning, and collaboration with QA Excellence on testing standards
- 2026 — Led Glass Decommissioning: delivered in 3 months what was estimated at 6 months with 4.5 FTEs, including unplanned scope — recognised with the adidas Bolder Award (April 2026); led go-live security process (security architecture review, penetration testing coordination); leading AI-driven creation and migration of end-to-end tests across Glass components
- Drove Latte cross-platform initiative: built the microfrontend from scratch and expanded JSON-driven rendering to Upper Funnel surfaces — checkout, PDP, PLP — coordinating joint API contracts and releases across .com and iOS/Android teams
- Internalized a fully external team (12 Merkle contractors → in-house) in 7 months; delivered the resulting microfrontend migration in 3 months on a hard deadline, extending the framework to support nested microfrontends
- Presented microfrontend migration strategy and Latte case study to ~100 adidas engineers each
- Defined and owned testing strategy: coverage thresholds, rotating quality champion role, Sonar gates, githooks, and improved deployment flows — lifting quality culture across both chapters

React · TypeScript · Node.js · XState v5 · AWS Lambda · DynamoDB · Kubernetes · Next.js · Latte

### Tech Lead & Sole Engineer

2024 (parallel to adidas)

Partum · Remote

- Designed and built a new generation smart locker kiosk from scratch — edge-first, Raspberry Pi + NestJS + React inside Docker managed by Balena
- Integrated Kerong smart lock hardware via a custom TCP binary protocol (STX/ETX framed) over raw sockets
- Kiosk UI driven entirely by XState v5 state machines (delivery and pickup flows)
- Moved cloud communication to AWS IoT and WebSockets; provisioned all infrastructure with AWS CDK v2

NestJS · TypeScript · React · XState v5 · Balena · AWS CDK v2 · Raspberry Pi

Software Engineer (Backend Lead, Connection Studio)

Jan 2019 – Aug 2022

adidas · Zaragoza, Spain

- Led the full re-architecture of Connection Studio 2.0: migrated from Java/Spring to Node.js/NestJS/TypeScript, replaced Oracle DB with DynamoDB — serving global adidas.com login and campaign traffic at scale
- Built a suite of internal NestJS libraries before equivalent open-source solutions existed: fluent DynamoDB query builder, decorator-driven Kafka router, @Cache() method decorator, runtime decorator discovery
- Reduced monthly deployments to weekly via GitHub Flow and CI/CD improvements; introduced Sonar, Instana, Grafana, Opsgenie
- Migrated mission-critical microservices to AWS Lambda; introduced Terraform and AWS CDK for IaC
- Security Champion: coordinated security architecture reviews and penetration testing for go-live; led security practices across the team throughout the re-architecture
- Mentored one engineer across 3+ years and 3 technology shifts (Java BE → Node.js BE → React FE)
- Promoted to Senior Software Engineer in 2022

TypeScript · Node.js · NestJS · AWS Lambda · DynamoDB · Apache Kafka · Terraform

Software Engineer

Jan 2014 – Jan 2019

GFT Group · Zaragoza, Spain (client: Deutsche Bank UK)

- Technical area lead for architecture, code quality, and security on the EUS project at Deutsche Bank UK
- 80% reduction in on-call incidents through Java and Unix quality engineering
- Led international knowledge transfer to HCL India (technical and functional, delivered in English); led disaster recovery planning across geographies
- Voluntarily took on frontend work on dbPalace after self-studying Angular when no frontend engineer was available

Java · Spring Boot · Angular · PL/SQL

Consultant / Analyst

May 2011 – Jan 2014

evis · Madrid, Spain (client: Ministerio de Economía y Competitividad)

- Developer and analyst for the unified application processing system and its data model
- Continuous development and improvement of the document processing system
- Created and improved Oracle stored procedures for inter-application data transfers

Java · Oracle SQL · PL/SQL

Researcher

Jan 2009 – Jan 2010

Universidad de Zaragoza · Zaragoza, Spain

- Research on Deep Belief Networks applied to brain-computer interfaces: using EEG signals to control a wheelchair through pattern recognition
- Implemented learning algorithms in Matlab; built a handwritten digit recogniser via camera as proof of concept
- Publication: [zaguan.unizar.es/record/5100](http://zaguan.unizar.es/record/5100)

Matlab · Deep Belief Networks · EEG signal processing

AWARDS

Bolder Award — AI Native Guild

adidas · April 2026

Selected by the M1 panel for the inaugural Bolder Awards for leading the Glass Decommissioning initiative. Estimated at 6 months with 4.5 FTEs — delivered in 3 months including unplanned scope, by pioneering GenAI-assisted component re-implementation. Built an AI-powered SDLC orchestration platform (Claude Agent SDK + XState v5 + Next.js) that drives tickets autonomously through analysis → implementation → code review → security audit → PR. Learnings shared across DXP and CE teams to drive broader AI adoption.

EDUCATION

Universidad de Zaragoza

2000 – 2010

Ingeniería Superior de Telecomunicaciones · Telecommunications Engineering — Master's equivalent (EQF Level 7)

LANGUAGES

Spanish Native

English C2 (Cambridge Grade A)

German Basic

SKILLS

LANGUAGES & FRAMEWORKS  
TypeScript · JavaScript · Node.js · React · Go · Python · Java · NestJS · XState · Express.js

CLOUD & INFRASTRUCTURE  
AWS (Lambda, DynamoDB, API Gateway, S3, CDK, IoT) · Kubernetes · Docker · Terraform · Balena · Apache Kafka · WebSockets · PostgreSQL · Instana · Grafana · Sonar

METHODOLOGIES  
Scrum (PSM I) · SAFe (Agilist + Practitioner) · Kanban · GitHub Flow · Microfrontend architecture

CERTIFICATIONS

Oracle Cloud Infrastructure 2023 Certified Architect Associate

Oracle · 2023

## AWS Certified Solutions Architect – Associate

AWS · 2022

## CKAD: Certified Kubernetes Application Developer

The Linux Foundation · 2022

## AWS Certified Developer – Associate (959/1000)

AWS · 2020

## CCA Spark and Hadoop Developer

Cloudera · 2019

## Certified SAFe® 4 Agilist

Scaled Agile · 2019

## Professional Scrum Master (PSM I)

Scrum.org · 2018

## Certified SAFe® 4 Practitioner

Scaled Agile · 2018

## Oracle Certified Professional, Java SE 8

Oracle · 2017

## Cambridge English: Advanced (CAE) — Grade A — C2

University of Cambridge · 2016

## PROJECTS

### Latte template AI analysis pipeline In progress

Using AI to build agents, commands, and tools that reduce tech debt across Latte templates — surfacing and fixing redundant state patterns, dead content, and hardcoded strings, while driving harmonisation across templates. Goal: lower the cost of creating new templates for markets, reduce the maintenance burden on existing ones, and reduce the risk of production issues caused by misconfiguration.

Claude Code TypeScript Next.js Latte

### AI homelab sandbox Active

Proxmox NAS running Ollama, n8n, and Claude in a sandboxed environment for testing AI integrations before bringing them to production. Used to explore advisor patterns, token reduction strategies, prompt injection hardening, and agentic workflow design.

Proxmox TrueNAS Ollama n8n Docker

### Home automation & IoT fleet Active

Declarative management of OpenBeken/Tuya smart devices via Ansible and Home Assistant. Being replaced by obk-fleet — a Go daemon using the Kubernetes operator pattern (desired state, reconciliation loop, drift detection) for full fleet management.

Go Ansible Home Assistant MQTT OpenBeken

### Technical writing In progress

Article series on AI for engineers — 'The Good, the Bad and the Ugly' — covering practical AI adoption, honest trade-offs, and engineering perspective on LLMs. Published on Medium and LinkedIn.

AI LLMs GenAI